

SRNN Ablation Variants

All variants are defined in `SRNN_PRESETS` in `train_srnn/models/srnn_cell.py` and as YAML configs in `conf/model/srnn_*.yaml`.

Parameter Reference

Parameter	Description
n_a_E	SFA timescales for excitatory neurons (0=off, 1=single, >=2=multi-timescale)
n_a_I	SFA timescales for inhibitory neurons
n_b_E	STD for excitatory neurons (0=off, 1=on)
n_b_I	STD for inhibitory neurons
dales	Dale’s law constraint (E columns positive, I columns negative)
echo	Reservoir/echo-state mode (freeze recurrent weights W)
per_neuron	Per-neuron SFA/STD time constants vs per-population
solver	ODE solver (semi_implicit, explicit, rk4, exponential)

Default values (from `SRNNConfig`): `n_a_E=3`, `n_a_I=3`, `n_b_E=1`, `n_b_I=1`, `dales=True`, `echo=False`, `per_neuron=False`, `solver=semi_implicit`.

Variant Table

Preset	n_a_E	n_a_I	n_b_E	n_b_I	dales	echo	per_neuron	solver	Description
srnn	3	3	1	1	Y	N	N	semi_implicit	Full model (all defaults)
srnn-per-neuron	3	3	1	1	Y	N	Y	semi_implicit	Per-neuron time constants
srnn-echo	3	3	1	1	Y	Y	N	semi_implicit	Reservoir mode (W frozen)
srnn-no-adapt	0	0	0	0	Y	N	N	semi_implicit	No SFA, no STD
srnn-no-adapt-no-dales	0	0	0	0	N	N	N	semi_implicit	No SFA, no STD, no Dale’s law
srnn-sfa-only	1	1	0	0	Y	N	N	semi_implicit	SFA only (1 timescale each)
srnn-std-only	0	0	1	1	Y	N	N	semi_implicit	STD only
srnn-E-only	3	0	1	0	Y	N	N	semi_implicit	Adaptation on E neurons only
srnn-e-only-echo	3	0	1	0	Y	Y	N	semi_implicit	E-only + reservoir mode
srnn-e-only-per-neuron	3	0	1	0	Y	N	Y	semi_implicit	E-only + per-neuron time constants
srnn-multi-sfa	2	2	1	1	Y	N	N	semi_implicit	2 SFA timescales (E and I)
srnn-multi-sfa-E	2	0	1	0	Y	N	N	semi_implicit	2 SFA timescales (E only) + STD E

Preset	n_a_E	n_a_I	n_b_E	n_b_I	dales	echo	per_neuron	solver	Description
srnn-no-dales	3	3	1	1	N	N	N	semi_implicit	Full model without Dale's law
srnn-explicit	3	3	1	1	Y	N	N	explicit	Full model, forward Euler solver
srnn-rk4	3	3	1	1	Y	N	N	rk4	Full model, RK4 solver

Batched Ablation Usage

Multiple variants can be run in parallel via BatchedSRNNCell using torch.bmm:

```
python train.py task=har batched_ablations='[srnn,srnn-no-adapt,srnn-E-only]'
```

All variants in a batch must share the same solver, h, and ode_unfolds. The solver-specific presets (srnn-explicit, srnn-rk4) cannot be mixed with the default semi_implicit variants in a single batch.

Visualization

```
# Compare two variants in detail (6 panels each)
PYTHONPATH=. .venv/bin/python scripts/compare_ablations.py

# Survey all 13 non-solver variants (3 panels each, wrapped grid)
PYTHONPATH=. .venv/bin/python scripts/compare_all_ablations.py --N 300
```